

Enis Furkan Kırmızı

MACHINE LEARNING · COMPUTER VISION · NLP · DATA ENGINEERING

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Artificial Intelligence & Data Engineering graduate who builds machine-learning systems end to end, from model training and compression to retrieval-augmented search and large-scale data pipelines. Senior thesis compressed neural networks by up to 64× while keeping the best configurations within 1.5% of the original model's accuracy; equally comfortable owning the path from raw data to trained model to a deployed service, with a focus on measurable, reproducible results.

EDUCATION

Istanbul Technical University

Istanbul, Türkiye

B.Sc. in Artificial Intelligence & Data Engineering | GPA 3.33 / 4.00

2021 - 2026

Relevant Coursework: Deep Learning · Natural Language Processing · Computer Vision · Data Mining · Analysis of Algorithms · Data Structures

TECHNICAL SKILLS

Languages: Python, C++, SQL

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers; knowledge distillation, model compression

Computer Vision & NLP: Vision Transformers, CNNs, image segmentation, multi-task learning; retrieval-augmented generation (RAG), embeddings, semantic search, rerankers

Data & Infrastructure: Apache Spark, Hadoop, Apache Iceberg, Airflow, PostgreSQL, vector databases, Docker, FastAPI, REST APIs, Git, AWS

EXPERIENCE

Central Securities Depository of Türkiye

Istanbul, Türkiye

AI & Data Platforms Intern

Jul 2026 – Aug 2026

- Developed and optimized **retrieval-augmented generation (RAG)** systems within the Data Analytics Application Development team to improve enterprise document retrieval and querying accuracy.
- Building and maintaining large-scale **data pipelines** that streamline high-volume data processing for downstream analytics workflows.

Ziraat Technology

Istanbul, Türkiye

Artificial Intelligence Intern

Aug 2025 – Sep 2025

- Built a **multimodal RAG pipeline** using CLIP to align text and image embeddings in a shared space, enabling context-aware retrieval over mixed text-image documents.
- Engineered an automated **document-ingestion pipeline** with OCR that extracts, chunks, and embeds PDF content into a vector database for semantic search.
- Implemented a **real-time face-recognition system** that detects faces and matches them using deep-learning face embeddings for live video identification.

PROJECTS

Joint Knowledge Distillation & Neural-Network Compression

B.Sc. Senior Thesis

Advised by Prof. Dr. Behçet Uğur Töreyn · PyTorch · Deep Learning · Model Compression

- Designed a framework that compresses convolutional neural networks and distills knowledge from a larger “teacher” model in a single training pass, achieving up to **64× smaller models** and 52× fewer computations while the best configurations stay within **1.5% of the teacher's accuracy** across 8 model pairs on CIFAR and ImageNet.
- Automated the tuning this kind of compression usually requires: the system sets its own per-layer compression rates and balances several training objectives with no hand-tuned coefficients.
- Applied **tensor decomposition** together with distributed, mixed-precision training, cutting CPU inference time **2.1×** (from 266 ms to 126 ms) and benchmarking several knowledge-distillation strategies.

Enterprise RAG Search Platform

Python · FastAPI · Vector Search · Docker

- Built an enterprise **retrieval-augmented generation (RAG)** system that enforces document-level access control directly in the database query, so users only retrieve documents they are authorized to see. Verified with **zero permission leaks across 480 user-query tests**.
- Combined **semantic vector search** with keyword search and a reranking model to return highly relevant results; **0.97 mean reciprocal rank**, with the correct document ranked first **95%** of the time, at 57 ms.
- Chose the embedding and reranking models through a reproducible evaluation harness, and covered the system with **75 automated tests** behind a continuous-integration access-control check.

Large-Scale Data Lakehouse

Apache Spark · Airflow · PostgreSQL · Cloud Storage

- Built a large-scale **data lakehouse** on Apache Spark with a layered (bronze/silver/gold) design that made analytical queries over **20 million records** more than **1,500× faster**; from about 15 seconds to roughly 10 milliseconds on the pre-aggregated layer.
- Added an **Airflow-orchestrated data-quality gate** that reconciles over **10 million records** against the source system before publishing, blocking bad data; validated at **zero discrepancies**.
- Enabled **point-in-time data recovery** (restoring 20 million rows in about 5 seconds) and schema changes with no data rewriting, while sustaining **200+ queries per second** under heavy concurrent load.

Multilingual Multimodal Idiom Understanding (AdMiRe)

Multimodal Machine Learning · NLP · Computer Vision

- Built a **multimodal (image + text)** system that ranks images by the figurative meaning of a phrase across **15 languages (13 zero-shot)**, scoring **0.37 top-1** on the official blind test versus a 0.28 baseline, all on a single 12 GB GPU.
- Chained machine translation, a language model for interpreting idioms, and image-text similarity matching into one pipeline, refined over six rounds of experiments.

Fine-Tuning SAM 2 for Video Object Segmentation

Computer Vision · PyTorch · DAVIS Benchmark

- Owned the data curation and preprocessing for a project fine-tuning Meta's **SAM 2** model for **video object segmentation** (tracking and outlining objects across video frames) evaluated on the DAVIS benchmark against a strong zero-shot baseline.
- Built the augmentation pipeline that broadened the training data to improve the model's generalization across varied video scenes.

Age & Gender Prediction from Facial Images

Computer Vision · PyTorch · Explainable AI

- Built a **multi-task** deep learning model (Vision Transformer and ResNet) that predicts both a person's age and gender from a single face image, sharing a common backbone across the two tasks for efficiency.
- Added explainability through attention maps and feature visualizations that show what drives each prediction.

Distributed Big-Data Analytics Pipeline

Apache Spark · Hadoop · Google Cloud

- Processed a **40 GB** dataset of user reviews with a distributed Apache Spark and Hadoop pipeline on Google Cloud, benchmarking single- vs multi-node clusters to measure the speedups from distributed, in-memory processing.
- Analyzed the data with sampling-based methods to profile user engagement and behavior trends efficiently at scale.

Additional Projects

- **Core Banking API**; a banking backend handling accounts and transactions · **Agora**; a Reddit-style social platform backend · **E-Commerce App**; an online store with AI-enhanced product search.